

HEALTHCARE ANALYTICS PLATFORM

Executive Portfolio Summary

Business Intelligence Portfolio Project

PostgreSQL • SQL • Tableau Desktop

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PROJECT OVERVIEW

The Healthcare Analytics Platform is an end-to-end Business Intelligence portfolio project designed to transform structured healthcare data into consistent, interactive, and decision-oriented analytical information.

The solution combines PostgreSQL and SQL for relational data management and centralized analytical preparation with Tableau Desktop for analytical calculations, KPI development, visualization, dashboard design, and interactive exploration.

The final analytical environment includes five executive dashboards covering Comparative Analytics, Revenue Distribution, Patient Demographics, Clinical & Medical Conditions, and Healthcare Operations. Together, these dashboards provide complementary financial, demographic, clinical, comparative, and operational perspectives from a common analytical foundation.

The project was developed using an educational healthcare dataset and represents a professional portfolio analytical implementation rather than a production enterprise deployment.

ARCHITECTURE & TECHNOLOGIES

The solution follows a layered Business Intelligence architecture in which PostgreSQL provides the structured data environment and centralized analytical preparation, while Tableau Desktop provides the analytical, visualization, interaction, and dashboard presentation capabilities.

The architecture is centered on the PostgreSQL database **healthcare_db** and the centralized analytical View **vw_healthcare_master**. Tableau consumes this standardized structure through **DS_vw_healthcare_master (healthcare_db)**, providing a common analytical data foundation for the five executive dashboards.

Tableau extends the centralized PostgreSQL foundation through Calculated Fields, Level of Detail (LOD) Expressions, Table Calculations, Sets, Parameters, ranking, segmentation, percentage calculations, time-based analysis, standardized KPIs, and interactive analytical capabilities.

Technology Stack: PostgreSQL • SQL • Tableau Desktop

Healthcare Data → PostgreSQL healthcare_db → vw_healthcare_master → Tableau Analytics → KPIs & Worksheets → Executive Dashboards

KEY ANALYTICAL CAPABILITIES

The Healthcare Analytics Platform demonstrates a broad range of Tableau analytical capabilities built on a centralized PostgreSQL data foundation. Reusable analytical logic supports KPI standardization, comparative analysis, segmentation, ranking, percentage contribution, and time-based analysis across the executive dashboard environment.

Key capabilities include:

- Calculated Fields and Level of Detail (LOD) Expressions
- Table Calculations, including ranking, percentages, Running Totals, and Moving Averages
- Sets and Parameters for segmentation and dynamic analytical behavior
- Selected versus Others comparative analysis
- Standardized KPI calculations
- Interactive filters and Dashboard Filter Actions for focused investigation
- Highlight Actions for contextual relationship analysis
- Trend, contribution, demographic, clinical, financial, and operational analysis

FIVE EXECUTIVE DASHBOARDS

The final Healthcare Analytics Platform includes five executive dashboards, each focused on a distinct analytical perspective while sharing the same centralized PostgreSQL and Tableau analytical foundation.

Comparative Analytics Dashboard — Supports Selected versus Others analysis to compare selected populations with the remaining dataset across revenue, patient counts, and related analytical measures.

Revenue Distribution Dashboard — Examines revenue distribution across medical conditions using KPIs, ranking, percentage contribution, and comparative financial analysis.

Patient Demographics Dashboard — Analyzes patient characteristics through age, gender, blood type, age-group composition, and insurance-provider perspectives.

Clinical & Medical Conditions Dashboard — Provides condition-level and clinical analysis across patient populations, medical conditions, medications, test results, and related healthcare attributes.

Healthcare Operations Dashboard — Evaluates admissions, discharges, billing activity, length of stay, admission types, insurance-provider revenue, test results, and monthly operational trends.

TECHNICAL SKILLS DEMONSTRATED

The Healthcare Analytics Platform demonstrates practical Business Intelligence competencies across database preparation, analytical development, visualization, dashboard design, interaction, validation, and technical documentation.

- **PostgreSQL & SQL** — relational data management, querying, centralized analytical preparation, and reusable View design
- **Tableau Desktop** — Calculated Fields, LOD Expressions, Table Calculations, Sets, Parameters, KPIs, worksheets, and dashboards
- **Analytical Methods** — comparative analysis, segmentation, ranking, percentage contribution, Running Totals, Moving Averages, and time-based analysis
- **Dashboard Development** — financial, demographic, clinical, comparative, and operational analytics
- **Interactive Analytics** — filters, Dashboard Filter Actions, and Highlight Actions
- **BI Architecture** — separation of data preparation, analytical logic, visualization, interaction, and presentation responsibilities
- **Validation & Documentation** — analytical validation, architectural documentation, and professional portfolio presentation

Portfolio Focus: Healthcare Data Analytics • Business Intelligence • Tableau Development • SQL Analytics